Multiple Linear Regression on Diabetes Progression

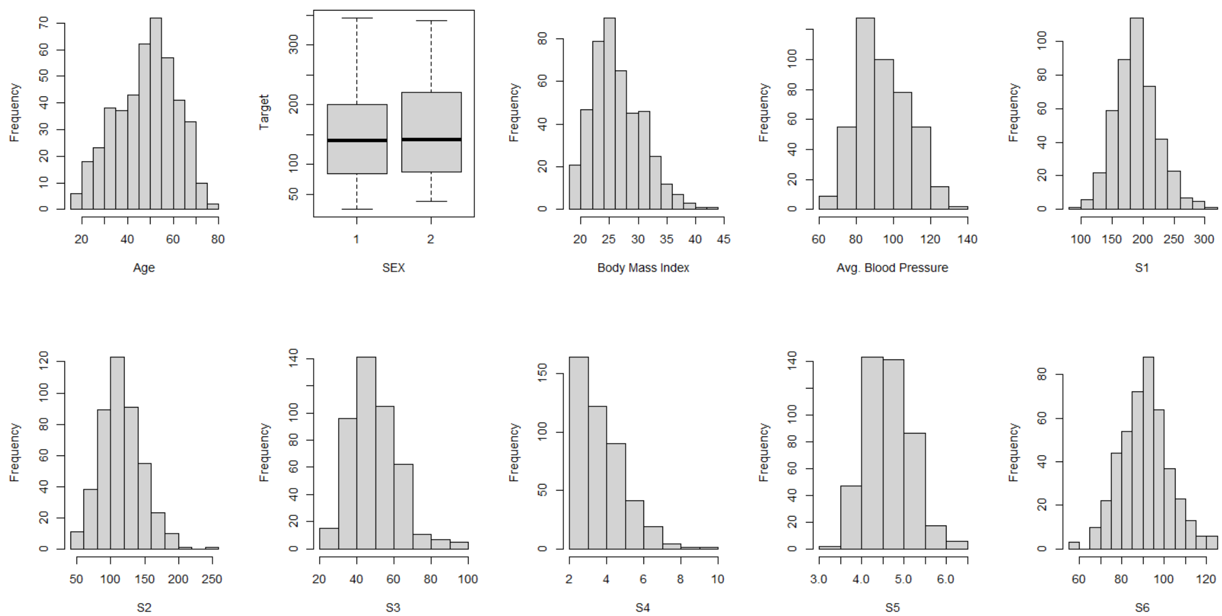
Alyssa Mercado

In this report I will show how I performed a regression analysis on the variation of diabetes progression based on 10 different observations. I am doing this to find the predictor variables that seem like they significantly affect the data, and show that they are relevant to the progression of diabetes. The dataset I'm using in the analysis includes variables on age, sex, body mass index (BMI), average blood pressure (BP), and 6 blood serum measurements (S1-S6). The 6 blood serum measurements are for serum total cholesterol level, low-density lipoproteins (LDL), high-density lipoproteins (HDL), total cholesterol/HDL ration, serum triglyceride level, and blood sugar level in that order. There is also a target variable, called Target, that is the disease progression indicator or response of interest I am basing the model off of. After checking I can see the given dataset has no missing values and has a total of 442 points of data, those being the statistics of diabetes patients.

To determine the distribution of the possible predicter variables, I will look at the summary statistics of the data as well as the visualizations of them with histograms and a boxplot. The five-number summary table below shows us the minimum, maximum, median, mean, the 1st, and the 3rd quarters of each of the variables individually. There is also a smaller chart under that which gives us the standard deviations for all of them as well. Using the summary though we can see that the youngest patient is 19 and the oldest is 79 years old, with the average age being 48 (and a variation of about 13 years between patients). Something to note is that the observation on sex is a binary variable, with 1 being male and 2 being female. There is moderately more male than female participants here, with a difference of 28 people. For BMI, the patients have an average of 26.38, with a low of 18 and a high of 42.20. The BMI for them has a standard deviation of 4.4, which is the variation. Average blood pressure ranges from 62-133 with a mean of 94.65, and has a standard deviation of 13.8. The first of the blood serum measurements S1, the serum total cholesterol level in patients, has a mean of 189.1 with patients having minimum of 97.0 and a maximum of 301.0. The statistics for S2, the number of low-density lipoproteins, shows that patients have a minimum of 41.60 and a maximum of 242.40 LDL's with a similar deviation to S1 at 30.4 (while S1's is 34.6). S3, the high-density lipoproteins variable, ranges between 22.00 to 99.00, with most patients having an average of 49.79 HDL's with a possible variation of about 12.9. The total cholesterol or HDL ratio (S4) has a low of 2.00 and a high of 9.09, with most patients having a 4.07 HDL level. The serum triglyceride levels of the patients (S5) have a small variation of 0.5 and an average of 4.64. The last possibly significant variable is S6 or the blood sugar level, which has a possible variation of 11.49 and an average of 91.26. Patients had a minimum blood sugar level of 58.00 and a maximum level of 124.00.

```
      AGE          SEX          BMI              BP
 Min.   :19.00   1:235   Min.   :18.00   Min.   : 62.00
 1st Qu.:38.25   2:207   1st Qu.:23.20   1st Qu.: 84.00
 Median :50.00           Median :25.70   Median : 93.00
 Mean   :48.52           Mean   :26.38   Mean   : 94.65
 3rd Qu.:59.00           3rd Qu.:29.27   3rd Qu.:105.00
 Max.   :79.00           Max.   :42.20   Max.   :133.00
       S1               S2               S3               S4
 Min.   : 97.0   Min.   : 41.60   Min.   :22.00   Min.   :2.00
 1st Qu.:164.2   1st Qu.: 96.05   1st Qu.:40.25   1st Qu.:3.00
 Median :186.0   Median :113.00   Median :48.00   Median :4.00
 Mean   :189.1   Mean   :115.44   Mean   :49.79   Mean   :4.07
 3rd Qu.:209.8   3rd Qu.:134.50   3rd Qu.:57.75   3rd Qu.:5.00
 Max.   :301.0   Max.   :242.40   Max.   :99.00   Max.   :9.09
       S5               S6             Target
 Min.   :3.258   Min.   : 58.00   Min.   : 25.0
 1st Qu.:4.277   1st Qu.: 83.25   1st Qu.: 87.0
 Median :4.620   Median : 91.00   Median :140.5
 Mean   :4.641   Mean   : 91.26   Mean   :152.1
 3rd Qu.:4.997   3rd Qu.: 98.00   3rd Qu.:211.5
 Max.   :6.107   Max.   :124.00   Max.   :346.0

       AGE        BMI         BP         S1         S2
13.1090278  4.4181216 13.8312834 34.6080517 30.4130810
        S3         S4         S5         S6     Target
12.9342022  1.2904499  0.5223906 11.4963347 77.0930045
```

While the five-number summary and the standard deviation gives us a clearer view of the statistics, being able to see the data visually is also beneficial. The histograms allow us to see the normality and the skewness that certain observations may have that aren't as noticeable in the summaries. The boxplot for the sex of the patients is more convenient for the binary variable as there are only two options, male or female. Starting with the histogram for the age of the patients it looks almost like it follows the normal distribution but has a slight left skew, meaning the patients are on average older. The variables BMI, LDL (S2), HDL (S3), and total cholesterol (S4) are all noticeably right skewed, so most have a value close to their respective minimums. The average blood pressure and S5 both don't seem to follow normality but also don't have a strong skew. The serum total cholesterol level (S1) and blood sugar level (S6) looks like it follows the normal distribution and lacks a strong skew.

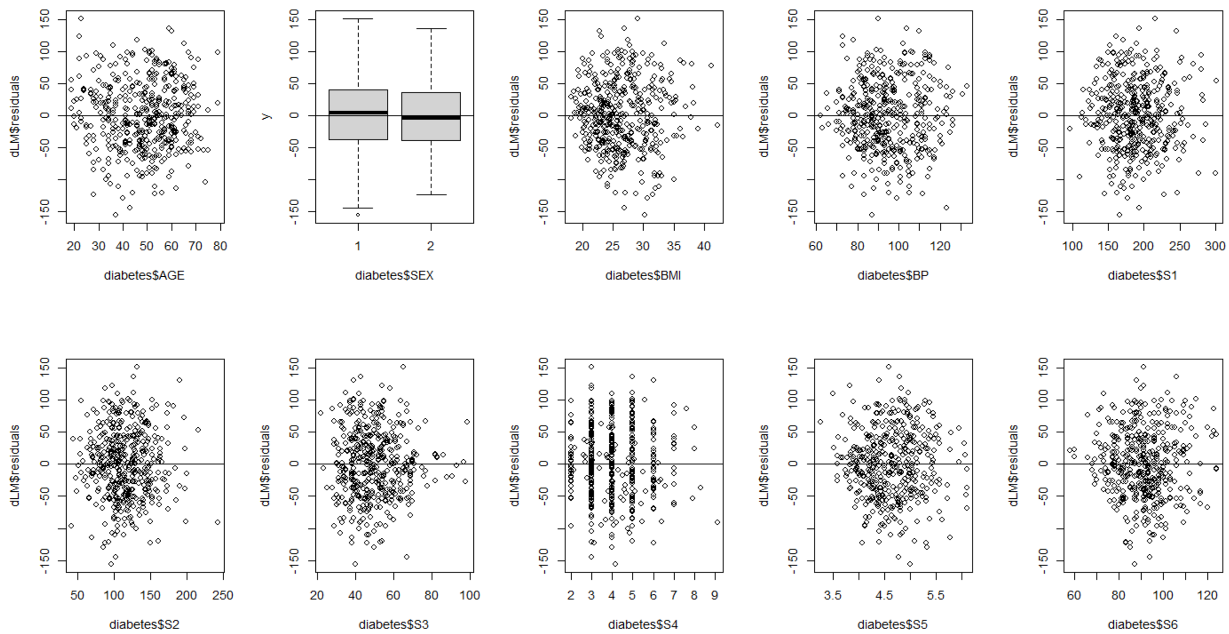


In the original model, we start with all 10 of the observations and base it off of the response variable. Looking first at its VIF values, we see that there’s an issue with multicollinearity for multiple variables (S1, S2, S3, S5). The VIF values are 59.2 for S1, 39.19 for S2, 15.40 for S3, and 10.07 for S5. To solve this issue, the later transformation will remove the S1 since it has the highest multicollinearity. When looking at the correlation between variables in a matrix, those same observations also have high values which confirms the potential issue with multicollinearity. Next, the ANOVA table will help determine which of the predictor variables significantly contributes to the variation in disease progression. The null hypothesis (for each observation) is that the variable does not significantly affect the response variable, while the alternative hypothesis is that it does. The p-values are how we decide this, having a very low one (usually <0.05) meaning that we must reject the null hypothesis and that the variable is significant. The table shows us that age, BMI, BP, S3, and S5 could be significant to explaining the response variable, while the other variables do not strongly contribute. After any necessary transformations and removals for collinearity they may be removed as well since they’re not needed. Underneath this is the summary statistics which provide information on the overall model, and the most relevant portion of those is the values for R squared and adjusted R squared. They're both quite similar, and using the normal value I can say that currently the model explains 51.77% of the variation in the progression of diabetes in these patients. The p-value of the F-statistic is also important because its close to 0, and similarly to the p-values from the ANOVA table this means that there are some predictors (likely the ones from the table) do contribute to the diabetes progression.

```
Analysis of Variance Table

Response: Target
           Df  Sum Sq Mean Sq  F value    Pr(>F)    
AGE         1   92527   92527  31.5504 3.490e-08 ***
SEX         1     293     293   0.1000    0.7519    
BMI         1  826955  826955 281.9792 < 2.2e-16 ***
BP          1  129312  129312  44.0934 9.448e-11 ***
S1          1    1791    1791   0.6108    0.4349    
S2          1    5058    5058   1.7246    0.1898    
S3          1  237329  237329  80.9257 < 2.2e-16 ***
S4          1    1821    1821   0.6210    0.4311    
S5          1   58856   58856  20.0690 9.582e-06 ***
S6          1    3080    3080   1.0504    0.3060    
```

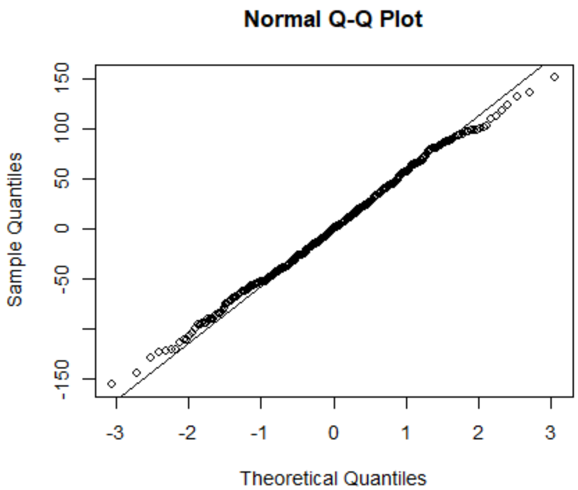
```
lm(formula = Target ~ AGE + SEX + BMI + BP + S1 + S2 + S3 + S4 + 
    S5 + S6, data = diabetes)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -357.42679   67.05807  -5.330 1.59e-07 ***
AGE           -0.03636    0.21704  -0.168 0.867031    
SEX2         -22.85965    5.83582  -3.917 0.000104 ***
BMI            5.60296    0.71711   7.813 4.30e-14 ***
BP             1.11681    0.22524   4.958 1.02e-06 ***
S1            -1.09000    0.57333  -1.901 0.057948 .  
S2             0.74645    0.53083   1.406 0.160390    
S3             0.37200    0.78246   0.475 0.634723    
S4             6.53383    5.95864   1.097 0.273459    
S5            68.48312   15.66972   4.370 1.56e-05 ***
S6             0.28012    0.27331   1.025 0.305990    
---

Residual standard error: 54.15 on 431 degrees of freedom
Multiple R-squared:  0.5177,	Adjusted R-squared:  0.5066 
F-statistic: 46.27 on 10 and 431 DF,  p-value: < 2.2e-16
```

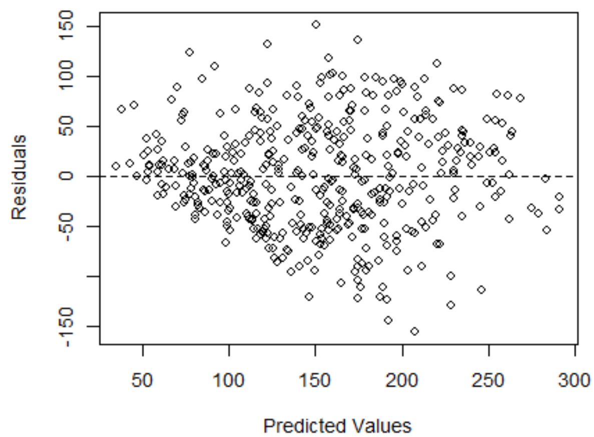
After reviewing the overall statistics of the model, I also need to make sure the assumptions of multiple linear regression are met. These include are linearity, independence, normality, and equal variance of the model which will be checked with several plots and tests. Independence is the only one not to be tested here because the data is not time-series. Beginning with linearity, plots of the data against the residuals of it need to have no pattern or be scattered. Overall it is followed by all the relevant variables except for the sex of the patient, since that one is binary and the vertical lines that are present in S4.



Normality and equal variance also need to be checked, and both have their own tests as well as plots. The null hypothesis is that the model follows the normal distribution, and the alternative is that it does not. This means we want the Shapiro-Wilk's test to not the reject the null hypothesis (p-value > 0.05) and for the q-q plot to follow the q-q line. We can see from the corresponding p-value/plot below that both of these seem true, and that the normality requirement is valid. The null hypothesis for our other assumption is that the model has equal variance, with the alternative being that it does not. Here we look at the Non-constant Variance test and the plot of the residual versus the predicted values. Both of these show the assumption is false since we reject the null hypothesis (p-value ~0) and since the plot has a fan like pattern. Since the model does not have equal variance we will have to do the Box-Cox transformation to make sure all of the requirements are met to have a properly fitted model.

```
Shapiro-Wilk normality test        Non-constant Variance Score Test
data:  dLM$residuals               Variance formula: ~ fitted.values
W = 0.99706, p-value = 0.6162      Chisquare = 11.1276, Df = 1, p = 0.00085052
```





As stated earlier, we first need to fix the multicollinearity issue by removing the variable with the highest VIF value, S1 to see if that changes the others. This removal does in fact fix the rest of values as they all end up being <10, thus removing the problem so we can continue to the next version of the model. Below are all of the current VIF's for each observation.

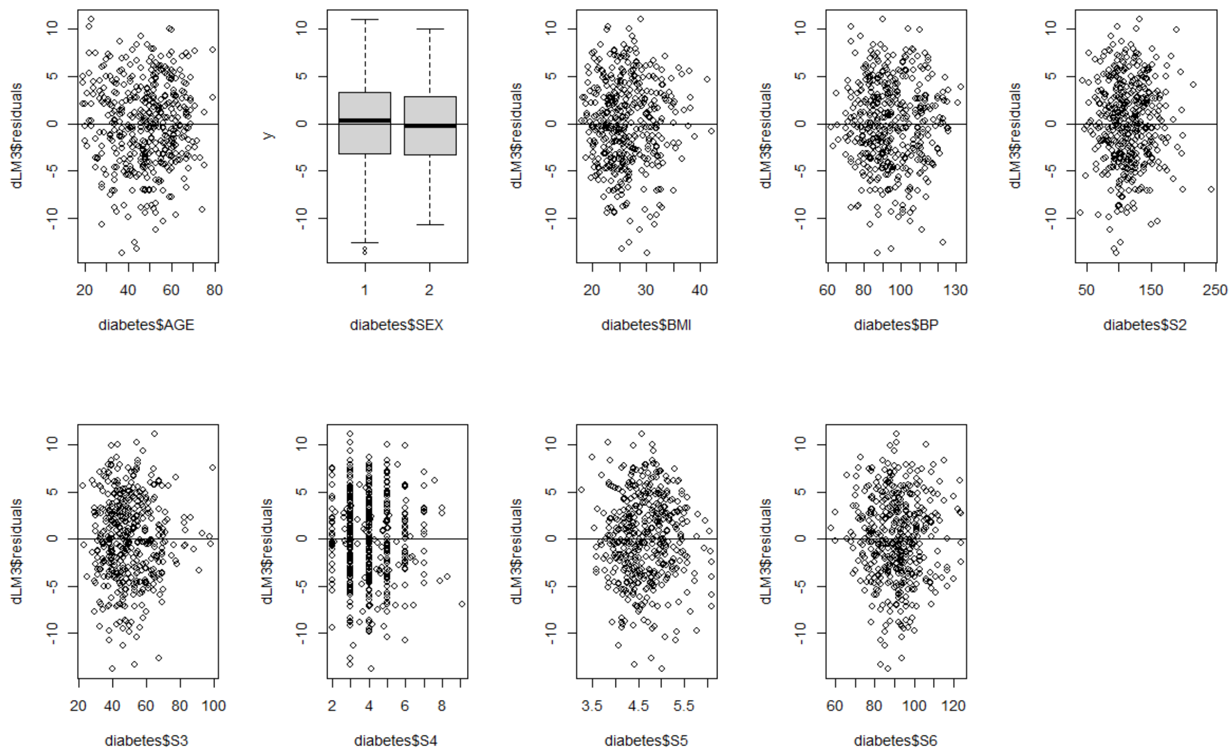
```
     AGE       SEX       BMI        BP        S2        S3        S4        S5        S6
1.216892  1.275049  1.502320 1.457413  2.926535  3.736890  7.818670  2.172865 1.484410
```

The model still does not follow the assumption of equal variance which can be fixed using the Box-Cox transformation. We do this by changing the response variable (Target) based on a new lambda, and then redoing the model and checking the summary/ANOVA table again. The null hypothesis here is still that the variable is significant, and the alternative is that it's not. Looking at the new table, the p-values have changed but age, BMI, BP, S3, and S5 continue to be possibly significant variables (since they're < 0.05). Now though S4's p-value has significantly decreased which means it could be relevant as well. The F-statistic continues to have a p-value close to 0, confirming that there continues to be observations that contribute to the diabetes progression, and the R squared has only slightly lessened by less than a percent.

```
Analysis of Variance Table

Response: TargetY
           Df Sum Sq Mean Sq  F value    Pr(>F)    
AGE         1  654.8   654.8  32.1011 2.677e-08 ***
SEX         1    1.3     1.3   0.0641 0.8003149    
BMI         1 5289.8  5289.8 259.3087 < 2.2e-16 ***
BP          1  835.6   835.6  40.9596 4.052e-10 ***
S2          1    8.6     8.6   0.4233 0.5156626    
S3          1 1009.2  1009.2  49.4707 7.927e-12 ***
S4          1  298.7   298.7  14.6420 0.0001492 ***
S5          1  856.2   856.2  41.9710 2.528e-10 ***
S6          1    9.3     9.3   0.4545 0.5005869    
```

```
lm(formula = TargetY ~ AGE + SEX + BMI + BP + S2 + S3 + S4 + 
    S5 + S6, data = diabetes)


Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.000e+01  3.062e+00  -3.266  0.00118 ** 
AGE          9.988e-04  1.810e-02   0.055  0.95602    
SEX         -1.887e+00  4.861e-01  -3.881  0.00012 ***
BMI          4.341e-01  5.967e-02   7.276 1.63e-12 ***
BP           8.718e-02  1.877e-02   4.644 4.54e-06 ***
S2          -9.356e-03  1.210e-02  -0.773  0.43971    
S3          -9.654e-02  3.214e-02  -3.003  0.00283 ** 
S4          -5.058e-02  4.660e-01  -0.109  0.91363    
S5           3.806e+00  6.069e-01   6.270 8.74e-10 ***
S6           1.537e-02  2.279e-02   0.674  0.50059    
---


Residual standard error: 4.517 on 432 degrees of freedom
Multiple R-squared:  0.5042,    Adjusted R-squared:  0.4939 
F-statistic: 48.82 on 9 and 432 DF,  p-value: < 2.2e-16
```

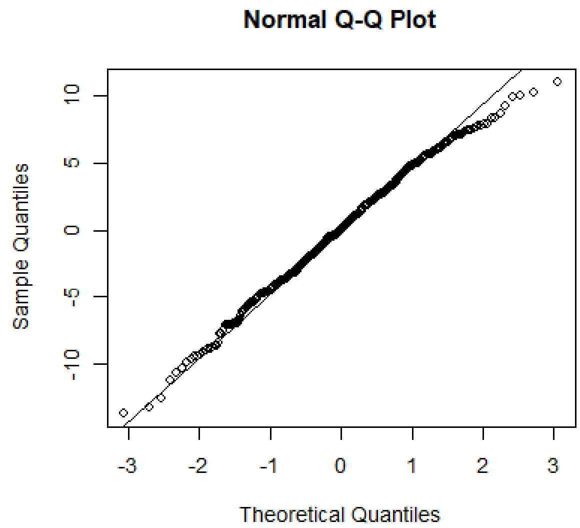
The Box-Cox transformation should’ve helped this new model now pass the equal variance test, but the other assumptions should also continue to be true. Starting again with linearity, not much has changed compared to the previous check. Since it was already confirmed to be linear this is the preferred outcome since the variance is what we wanted to have changed.

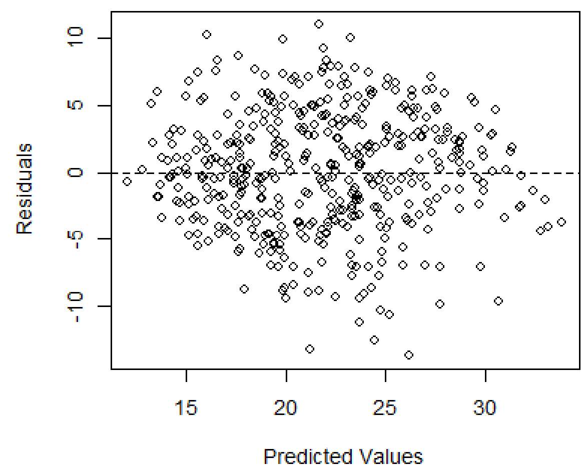


The model continues to follow the normal distribution, since the Shapiro-Wilks test does not reject the null hypothesis of normality and the q-q plot still follows the line properly. What has changed is the resulting p-value for the NCV test is no longer less than 0.05, so it does not reject the null hypothesis of equal variance. The corresponding graph of residuals versus predicted values now also does not have the fan pattern but is much more scattered, which was the goal of the Box-Cox transformation. Now that the model follows the L.I.N.E assumptions, we can start eliminating redundant variables in the final model.

```
Shapiro-Wilk normality test
data:  dLM3$residuals
W = 0.99423, p-value = 0.09337
```

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 1.096515, Df = 1, p = 0.29503
```





When creating the final model, I want to build the best one by using backward selection and beginning with the full model (which is currently the transformed one shown above). Using the summary information from that model I can see that age, S2, S4, and S6 have high p-values (> 0.10) and therefore need to be removed. The one with the highest p-value, age, was removed first and then I redid the summary statistics and ANOVA table to confirm it was not a significant variable. The table compares the full model to one without the age observation, and the high p-value shows that the removed variable does not explain a significant portion of the response (so it is not needed). The R squared has not changed though, and the model continues to explain 50.42% of the response variable Target.

```
Analysis of Variance Table
    Res.Df    RSS Df Sum of Sq     F Pr(>F)
1    432 8812.6
2    433 8812.7 -1 -0.062128 0.003  0.956

lm(formula = Target ~ SEX + BMI + BP + S2 + S3 + S4 + S5 + S6, 
    data = diabetes2)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -10.009430   3.053865  -3.278 0.001131 ** 
SEX          -1.884083   0.483150  -3.900 0.000112 ***
BMI           0.434118   0.059598   7.284 1.54e-12 ***
BP            0.087376   0.018411   4.746 2.82e-06 ***
S2           -0.009298   0.012037  -0.772 0.440292    
S3           -0.096464   0.032078  -3.007 0.002790 ** 
```

```
S4           -0.051153   0.465380  -0.110 0.912526    
S5            3.808767   0.603312   6.313 6.78e-10 ***
S6            0.015532   0.022568   0.688 0.491687    
---

Residual standard error: 4.511 on 433 degrees of freedom
Multiple R-squared:  0.5042,	Adjusted R-squared:  0.4951 
F-statistic: 55.05 on 8 and 433 DF,  p-value: < 2.2e-16
```

The next variable to be removed is S4 because it has the second highest p-value after the refitting. This time the ANOVA table compares the new model to the last one with the removed age variable, and we see that it once again has a high p-value. So, we are safe to remove this observation as well since it doesn't seem to change the model of our data or be significant to it. The R squared has not changed, but the adjusted R squared (which takes into consideration the number of observations we have) has slightly increased to 49.62% from 49.51%, also showing we are correct in removing this variable.

```
Analysis of Variance Table

    Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    433 8812.7                             
2    434 8812.9 -1   -0.2459 0.0121 0.9125

lm(formula = Target ~ SEX + BMI + BP + S2 + S3 + S5 + S6, data = diabet
es2)


Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -10.09254    2.95542  -3.415 0.000698 ***
SEX          -1.88758    0.48156  -3.920 0.000103 ***
BMI           0.43445    0.05946   7.307 1.32e-12 ***
BP            0.08756    0.01832   4.780 2.40e-06 ***
S2           -0.01032    0.00762  -1.355 0.176273    
S3           -0.09371    0.02005  -4.675 3.94e-06 ***
S5            3.77594    0.52361   7.211 2.48e-12 ***
S6            0.01536    0.02249   0.683 0.494940    
---

Residual standard error: 4.506 on 434 degrees of freedom
Multiple R-squared:  0.5042,	Adjusted R-squared:  0.4962 
F-statistic: 63.06 on 7 and 434 DF,  p-value: < 2.2e-16
```

The last summary showed that S6 has the next highest p-value, and therefore should be removed as well. The ANOVA table below compares it to the last model with two removed variables, and once again has a large p-value. S6 being removed did not significantly affect our model and also increased the adjusted R squared by 0.0006, meaning that the current model explains 49.68% of our response variable now.

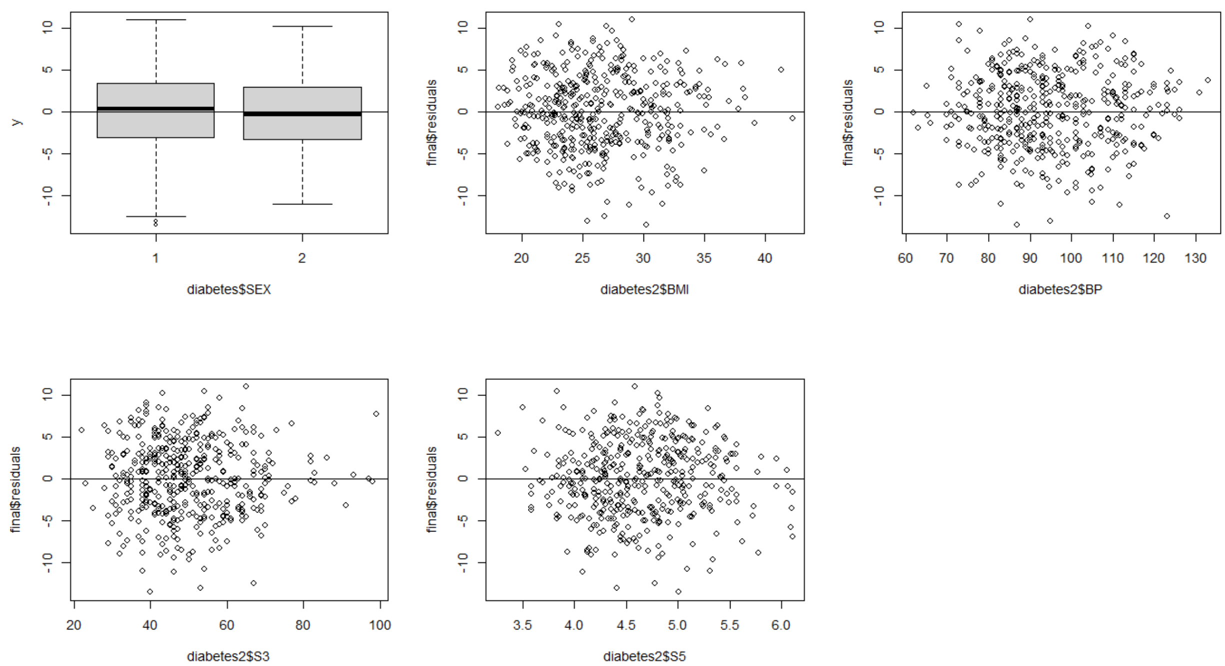
```
Analysis of Variance Table

  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    434 8812.9
2    435 8822.4 -1    -9.474 0.4666 0.4949


lm(formula = Target ~ SEX + BMI + BP + S2 + S3 + S5, data = diabetes2)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.561516   2.849585  -3.355 0.000862 ***
SEX2        -1.857189   0.479201  -3.876 0.000123 ***
BMI          0.440530   0.058748   7.499 3.66e-13 ***
BP           0.089836   0.017999   4.991 8.69e-07 ***
S2          -0.009650   0.007552  -1.278 0.201981
S3          -0.094043   0.020029  -4.695 3.57e-06 ***
S5           3.866248   0.506337   7.636 1.44e-13 ***
---

Residual standard error: 4.503 on 435 degrees of freedom
Multiple R-squared:  0.5037,    Adjusted R-squared:  0.4968
F-statistic: 73.58 on 6 and 435 DF,  p-value: < 2.2e-16
```

The last variable to be removed is S2, which has a p-value of 0.20 in the previous summary. In the ANOVA comparing it to the earlier model we see it has a p-value above 0.05 but it's not particularly large. This may be why the R squared did not improve much, but the variable itself still didn't contribute much to the response variable. This now becomes the final model to be tested for the multiple linear regression assumptions, and it explains 50.18% of the response variable.

```
Analysis of Variance Table

  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    435 8822.4
2    436 8855.5 -1   -33.118 1.6329  0.202

lm(formula = Target ~ SEX + BMI + BP + S3 + S5, data = diabetes2)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.79954    2.84555  -3.444 0.000629 ***
SEX2        -1.90693    0.47796  -3.990 7.76e-05 ***
BMI          0.43141    0.05836   7.393 7.42e-13 ***
BP           0.08948    0.01801   4.968 9.71e-07 ***
S3          -0.09361    0.02004  -4.671 4.00e-06 ***
```

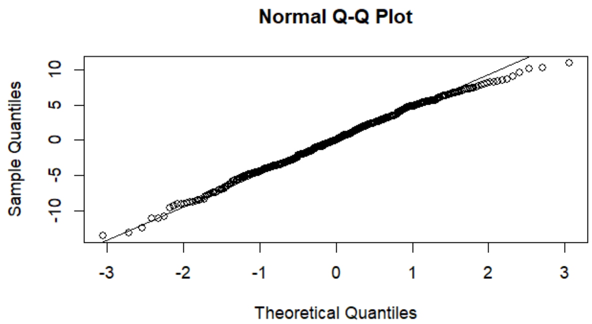
```
S5           3.73699    0.49649   7.527 3.01e-13 ***
---

Residual standard error: 4.507 on 436 degrees of freedom
Multiple R-squared:  0.5018,    Adjusted R-squared:  0.4961
F-statistic: 87.84 on 5 and 436 DF,  p-value: < 2.2e-16
```

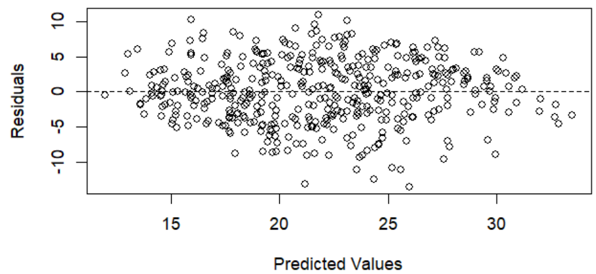
The final model now just needs to follow all the proper L.I.N.E assumptions. The linearity for the remaining variables is checked by the plots of sex, BMI, BP, S2 and S5. Each of the applicable plots has points scattered across and no discernible pattern which means that the final model does in fact pass the linearity assumption.



This model continues to follow a normal distribution, with the Shapiro Wilk's test having a p-value above 0.05 (do not reject the possibility of normality) and q-q plot properly following the q-q line as well. The Non-constant variance test gives us an even larger p-value, meaning it also passes and does not reject the null hypothesis of equal variance. We can see in the plot of residuals versus predicted values for the plot that it has no pattern, confirming the hypothesis.

```
Shapiro-Wilk normality test       Non-constant Variance Score Test
data:  final$residuals            Variance formula: ~ fitted.values
W = 0.99423, p-value = 0.09346   Chisquare = 0.9597226, Df = 1, p = 0.32726
```





In conclusion, this final model explains 50.18% of the variation in the progression of diabetes in these patients. I think that it is possible the R squared could be increased if there was even more relevant statistics that could be taken from the patients as well, or if there was a larger sample size. Generally, the R squared did not change much throughout the different models, with only small increases and decreases. The variables of sex, body mass index, average blood pressure, count of high-density lipoproteins, and serum triglyceride level are all significant in predicting the response variable. This means that each of these observations from the dataset seem to have a strong relationship with diabetes progression as shown in the ANOVA tables and summaries.

Appendix:

```
#################################################################################
############################# Diabetes Model ###################################
#################################################################################
#################################################################################

diabetes <- read.table("dataset/Diabetes.txt", header = TRUE)

diabetes$SEX = as.factor(diabetes$SEX)
sum(is.na(diabetes))
summary(diabetes)
sapply(diabetes[ , -2], mean)
sapply(diabetes[ , -2], sd)
par(mfrow = c(2,5))
hist(diabetes$AGE, xlab = "Age", main = "")
boxplot(Target ~ SEX, data = diabetes, main = "")
hist(diabetes$BMI, xlab = "Body Mass Index", main = "")
hist(diabetes$BP, xlab = "Avg. Blood Pressure", main = "")
hist(diabetes$S1, xlab = "S1", main = "")
hist(diabetes$S2, xlab = "S2", main = "")
hist(diabetes$S3, xlab = "S3", main = "")
hist(diabetes$S4, xlab = "S4", main = "")
hist(diabetes$S5, xlab = "S5", main = "")
hist(diabetes$S6, xlab = "S6", main = "")
```

```
################################################################################
############################# Original Model ###################################
################################################################################
################################################################################

dLM <- lm(Target ~AGE + SEX + BMI + BP + S1 + S2 + S3 + S4 + S5 + S6, data = diabetes)
anova(dLM)
summary(dLM)
library(car)
vif(dLM)
cor(diabetes[, c("AGE", "BMI", "BP","S1", "S2", "S3", "S4", "S5", "S6")])
# s1 has severe multicollinearity

######################################################## L.I.N.E assumptions

# linearity
par(mfrow = c(2,5))
plot(x = diabetes$AGE, y = dLM$residuals)
abline(h=0)
plot(x = diabetes$SEX, y = dLM$residuals, xlab = "diabetes$SEX")
abline(h=0)
plot(x = diabetes$BMI, y = dLM$residuals)
abline(h=0)
plot(x = diabetes$BP, y = dLM$residuals)
abline(h=0)
plot(x = diabetes$S1, y = dLM$residuals)
```

```
abline(h=0)

plot(x = diabetes$S2, y = dLM$residuals)

abline(h=0)

plot(x = diabetes$S3, y = dLM$residuals)

abline(h=0)

plot(x = diabetes$S4, y = dLM$residuals)

abline(h=0)

plot(x = diabetes$S5, y = dLM$residuals)

abline(h=0)

plot(x = diabetes$S6, y = dLM$residuals)

abline(h=0)


# normality, H0: normal, do not reject

par(mfrow = c(2,3))

qqnorm(dLM$residuals)

qqline(dLM$residuals)

shapiro.test(dLM$residuals)


# equal variances, H0: equal, reject

ncvTest(dLM)

plot(x = dLM$fitted.values, y = dLM$residuals, xlab = "Predicted Values", ylab = "Residuals")

abline(h = 0, lty = 2)


################################################ remove S1


dLM2 <- lm(Target ~AGE + SEX + BMI + BP + S2 + S3 + S4 + S5 + S6, data = diabetes)
```

```
vif(dLM2)

# This  elimates multico. in the rest of variables


########################################################################################

############################# Box Cox Model ########################################

########################################################################################

########################################################################################


############################################################# transformation

library(MASS)

trans <- boxcox(Target ~AGE + SEX + BMI + BP + S2 + S3 + S4 + S5 + S6,

        data = diabetes, plotit = F, lambda = seq(-3, 3, by = 0.125))

lambda <- trans$x[which.max(trans$y)] #  lambda value


## Add new column to data frame with transformed variable

diabetes$TargetY <- ((diabetes$Target^lambda) - 1) /lambda


########################################################## model after transformation


dLM3 <- lm(TargetY ~AGE + SEX + BMI + BP + S2 + S3 + S4 + S5 + S6, data = diabetes)

anova(dLM3)

summary(dLM3)


############################################################## L.I.N.E assumptions


# Linearity
```

```
par(mfrow = c(2,5))

plot(x = diabetes$AGE, y = dLM3$residuals)

abline(h=0)

plot(x = diabetes$SEX, y = dLM3$residuals, xlab = "diabetes$SEX")

abline(h=0)

plot(x = diabetes$BMI, y = dLM3$residuals)

abline(h=0)

plot(x = diabetes$BP, y = dLM3$residuals)

abline(h=0)

plot(x = diabetes$S2, y = dLM3$residuals)

abline(h=0)

plot(x = diabetes$S3, y = dLM3$residuals)

abline(h=0)

plot(x = diabetes$S4, y = dLM3$residuals)

abline(h=0)

plot(x = diabetes$S5, y = dLM3$residuals)

abline(h=0)

plot(x = diabetes$S6, y = dLM3$residuals)

abline(h=0)


# normality, H0: normal, do not reject

par(mfrow = c(2,3))

qqnorm(dLM3$residuals)

qqline(dLM3$residuals)

shapiro.test(dLM3$residuals)
```

```
# equal variances, do not reject
ncvTest(dLM3)
plot(x = dLM3$fitted.values, y = dLM3$residuals, xlab = "Predicted Values", ylab = "Residuals")
abline(h = 0, lty = 2)

#####################################################################################
############################# Full Model ######################################
#####################################################################################
#####################################################################################

diabetes2 <- data.frame(Target = diabetes$TargetY,
                        AGE = diabetes$AGE,
                        SEX = diabetes$SEX,
                        BMI = diabetes$BMI,
                        BP = diabetes$BP,
                        S2 = diabetes$S2,
                        S3 = diabetes$S3,
                        S4 = diabetes$S4,
                        S5 = diabetes$S5,
                        S6 = diabetes$S6)

full <- lm(Target ~., data = diabetes2)
anova(full)
summary(full)

############################################################## backward selection
```

```
# AGE, S2, S4, S6 have large p values


lmR1 <- lm(Target ~SEX + BMI + BP + S2 + S3 + S4 + S5 + S6, data = diabetes2)

anova(full, lmR1)

summary(lmR1)


lmR2 <- lm(Target ~SEX + BMI + BP + S2 + S3 + S5 + S6, data = diabetes2)

anova(lmR1, lmR2)

summary(lmR2)


lmR3 <- lm(Target ~SEX + BMI + BP + S2 + S3 + S5, data = diabetes2)

anova(lmR2, lmR3)

summary(lmR3)


lmR4 <- lm(Target ~SEX + BMI + BP + S3 + S5, data = diabetes2)

anova(lmR3, lmR4)

summary(lmR4)


###########################################################################################

################################## Final Model ##########################################

###########################################################################################

###########################################################################################


final <- lm(Target ~SEX + BMI + BP + S3 + S5, data = diabetes2)

summary(final)
```

```
anova(final)


# Linearity

par(mfrow = c(2,3))

plot(x = diabetes2$SEX, y = final$residuals, xlab = "diabetes$SEX")

abline(h=0)

plot(x = diabetes2$BMI, y = final$residuals)

abline(h=0)

plot(x = diabetes2$BP, y = final$residuals)

abline(h=0)

plot(x = diabetes2$S3, y = final$residuals)

abline(h=0)

plot(x = diabetes2$S5, y = final$residuals)

abline(h=0)


# normality, H0: normal, do not reject

par(mfrow = c(2,2))

qqnorm(final$residuals)

qqline(final$residuals)

shapiro.test(final$residuals)


# equal variances, do not reject

ncvTest(final)

plot(x = final$fitted.values, y = final$residuals, xlab = "Predicted Values", ylab = "Residuals")

abline(h = 0, lty = 2)
```